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Bidirectional Long Term Short Memory Based Text Mining Classification Method for Analysing the What's App Messages with Text and Emoji

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***Abstract*-- In this digital time, from plain text, emoji's are crucial to supplement emotional expression. However, effectively understanding and applying text mining techniques to messages containing both text and emoji's still continues to pose great challenges. This paper proposes a model based on Hybrid CNN-BiLSTM for enhanced emotion recognition for WhatsApp messages. The model utilizes CNN for the extraction of spatial features and BiLSTM for establishing the contextual relation of sequences to text. The framework begins with pre-processing of WhatsApp messages through tokenization of characters, filtering of stop words, normalizing emoji into meaningful representations. The CNN module extracts essential features from input sequences, while the BiLSTM works on learning the context dependency both into the past and into the future. The final classification layer implements a fully connected network with a soft-max activation for sentiment and emotional classification. The model was trained on a diverse and annotated WhatsApp dataset, in which text and emoji patterns are present, rendering it suitable for learning and generalisation tasks. Experimental results have demonstrated that the proposed method performs better than traditional machine-learning techniques in accuracy, precision, recall, and F1-score. The proposed system serves as a quick response to sentiment recognition via WhatsApp, surveillance of social media, and human-computer interaction, grasping the entire complexity of user emotions in digital communication. This study gives further impetus for emoji-aware text mining, providing substantial help within emotion recognition systems under natural language processing.**

***Keywords*: Emoji-to-Text conversion, Emotion recognition, BiLSTM algorithm, Text mining.**

I. Introduction

Text mining represents a more advanced analytical process whereby tangible value and deep meaning are extracted from unstructured interactions of text data. With the exponential increase in textual content sources-e.g., social media, blogs, news articles, and corporate documents-this area has gained enormous importance in recent times. Employing some techniques from [1], Natural Language Processing (NLP), machine learning, and data mining, text mining allows the identification of patterns, trends, and relationships that are inherently embedded in the text.

In modern data-driven existence, text mining is a key dot on the big picture. Organizations use techniques in text mining to compute the sentiments of their consumers, automate their data-processing tasks, and derive actionable knowledge from big-text datasets. The ability to get insights into text not only enhances operating efficiencies but provides additional advantage in this complex information environment.

They provide a strong communication ground of self-expression to its users who communicate [2], in any form of text or multimedia type. From all these forms, text has remained to dominate as the most frequently used means of self-expression. But, in recent times, emoji's have become meaningful and important to supplement text as a way to express feelings, actions, or reactions. Text has its own limitations in conveying emotions or the tone. To attenuate, users have taken recourse to adding a lot of emoji.

Emojis

Emoji's are little digital figures, representations, or images to convey emotions, objects, or ideas. Created in Japan in the late 1990s, they gained popularity in social-media as a form of global expression on Facebook, Twitter, Instagram, WhatsApp, etc. [3], Emoji's help in giving connotation, context, and emotional graces to mood variations in a neutral or ambiguous text, and help in better clarity of communication. Interpretations of emoji emotive significance when used as meaning-laden text is a challenging task, which can have applications in sentiment analysis, Chabot development, and user experience analytics and research.

Types of Emojis

Table 1 contains different types of Emojis. The table classifies emoji's into different classes and provides examples of each type. Emojis are always evolving and are used to illustrate the great diversity of ideas in digital communication. Even though they lighten digital communication, their presence in text analysis and Natural Language Processing is an impediment. Emojis can have multiple readings based upon the context, culture, and even their platform. The mere fact that transforming text into emoji's is not an easy task to achieve implies that any such algorithm must be sophisticated enough to capture both textual content and emotional context [4].

II. Related Works

Emoji can be considered a universal language. A study on their linguistic role in social messaging describes them as integral grammatical elements in digital communication on social media can be thought of as a universal or global language. Emoji, according to the authors in [5], are an evolution of visual language rather than a brand-new language, and they can add additional levels of meaning to digital communication. Emoji are used "to prevent misunderstandings and replace textual expressions," according to the authors of [6]. Emoji can occasionally take the role of words, according to a study on the usage of stickers and emoji by thirty WeChat users [7]. Advertisers frequently use emoji to boost positive affect and buy intention because they are entertaining and enticing [8].

Among the text extraction techniques, one of the hottest topics of interest has been the analysis of social media content owing to the immense rise of platforms like Twitter. The scholars have developed methods capable of classifying sentiments expressed through tweets into favourable, unfavourable, and neutral, thereby providing insights on public opinion and emotional responses. Chaudhary et al [11], integrated hashtag, Emoji, and n-gram combinations together allow for the finer granularity of emotional data extraction from text, helping classify tweets with great accuracy into groups of sentiment differences, usually working around a 92% accuracy mark. The increase in data generation is rising fast, along with the difficulties of the general interpretation of tweets in terms of their short and often obscure language.

The computers that deal with these problems create an open door for applications in public health and social monitoring, thereby expanding the usability of text extraction methodologies across varying domains.

Additionally, the development of a cyber-dictionary that would mine user-generated content would allow machine-learning algorithms to create predictive models that can reflect real-time linguistic trends (Hales et al.). In this way, these technologies can be used to gain insights into social conversations with practical applications in health and communication.

The next sections highlight the following: the existing methods are outlined in Section 3, while Section 4 discusses the proposed work in detail. Results and Discussion comparing experimental results with existing approaches is discussed in Section 4, while Section 5 concludes and provides Future Work.

III. Existing Methodology

A. Bidirectional-Long Short Term Memory

BiLSTMs are a more powerful tool for text mining, allowing for the descriptive modeling of complex contextual relationship within text data. The introduction of a bidirectional RNN serves to provide a better portrayal of syntactic, semantic, and pragmatic nuances of a given language, where the RNN reads the input sequences in both directions-left to right and right to left. These properties make BiLSTMs most effective for various tasks in sentiment analysis, named entity recognition, and part of speech tasks. A BiLSTM consists of two LSTMs, with one reading the sequence from left to right and the other reading the input from right to left; thus, being able to capture both local and long-range dependencies. The output of both LSTMs is combined to form one vector, which is then used for activation. BiLSTM models are particularly useful for text-mining applications requiring consideration of both local and global contexts.

B. Convolutional Neural Networks

CNNs have become indispensable in text mining, offering a powerful means of obtaining salient and constructive features of a text data. Although CNNs were initially developed for images, they have been naturally adapted to text analysis wherein each unit in the CNN finds one-dimensional text. They are also favored in this domain for their ability to learn local patterns and relations such as sentiments in phrases or the set of keywords regarding a subject. Through convolution and pooling layers, CNNs learn subjectively relevant features, dispensing with some of the meticulous feature engineering. CNNs can easily integrate with other types of deep learning architecture like RNN and LSTM, providing a further redoubling of strength for text analyses. In short, this indicates

Table 1: Types of Emojis

Category	Description	Examples
Face Emojis	Represent facial expressions, emotions, or reactions.	😊 (smiling), 😢 (crying), 😠 (angry), 😍 (heart-eyes)
Object Emojis	Represent physical objects or items.	🍕 (pizza), 📱 (smartphone), 🏠 (house), 🎸 (guitar)
Nature Emojis	Represent elements of nature and living things.	🌳 (tree), 🌞 (sun), 🦋 (butterfly), 🌧 (rain)
Symbol Emojis	Represent abstract concepts or symbolic representations.	♡ (heart), 🔥 (fire), ✨ (sparkles), 💯 (100)
Food Emojis	Represent various foods and beverages.	🍎 (apple), 🍔 (burger), 🍺 (beer), 🍩 (donut)
Animal Emojis	Represent animals and creatures.	🐶 (dog), 🦄 (unicorn), 🐯 (tiger), 🦉 (owl)
Activity Emojis	Represent activities or hobbies.	⚽ (soccer), 🎮 (video game), 🧘 (yoga), 🎨 (painting)
Transportation Emojis	Represent modes of transportation.	🚗 (car), 🚆 (train), ✈️ (airplane), 🚀 (rocket)

why CNNs become important and intensely useful in text mining, providing researchers and practitioners alike real insights or other hidden nuggets.

IV. Proposed Work

The conversion of text to emoji has found fame in text mining. The process of text-to-emoji mapping adds additional emotional cues to text communication. Prior techniques were mainly montaged on sentiment analysis and fixed symbolic rules for text-to-emoji mapping, showing nuances in contextually accurate emoji assignment. DeepEmojizer, on the other hand, offers a hybrid mode combining sequential processing strengths of BiLSTM with feature extraction capability of CNN, thereby providing a much sturdier and scalable system. The proposed method is illustrated in its overall flow chart in Fig. 1.

A.Text Pre-Processing

The proposed Hybrid CNN-BiLSTM Model is designed to improve text mining in WhatsApp messages by incorporating both text and emoji-based sentiment analysis. The method begins with data collection, where WhatsApp messages containing text and emojis are gathered for processing. Next, a preprocessing step is applied, which involves tokenization (splitting text into words), lowercasing (standardizing text), stopword removal (eliminating redundant words), and lemmatization/stemming (converting words to their base form). Since emojis play a crucial role in sentiment expression, they are extracted and normalized by mapping them to predefined emotional categories (e.g., 😀 → "happy", 😢 → "sad").

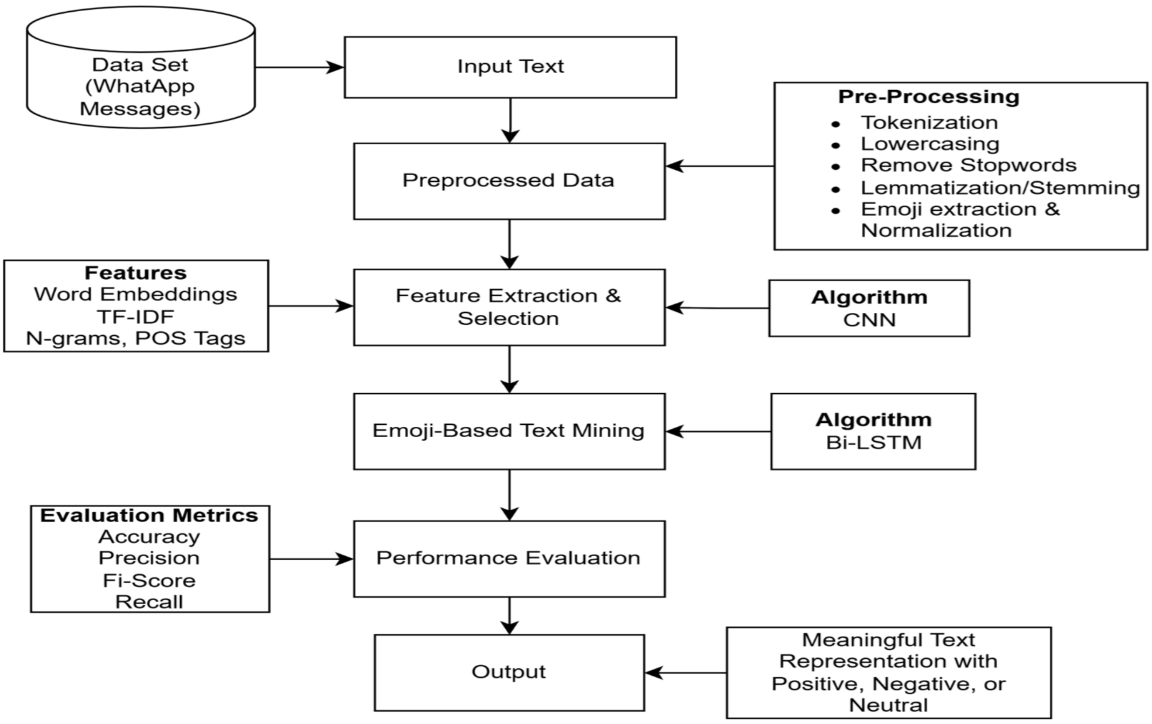

Fig 1. Proposed Methodology of Emoji-Based Text Mining

The processed text and emoji features are then fed into the hybrid CNN-BiLSTM model. The CNN component extracts spatial and local text patterns, while the BiLSTM component captures long-range dependencies in sequential data. These features are combined and passed through a fully connected neural network, followed by a softmax classifier to predict sentiment categories such as positive, negative, or neutral.

B.Feature Extraction using CNN

In the feature extraction phase of the proposed model, both text and emojis are transformed into numerical representations to enable deep learning-based sentiment analysis. First, the textual content of WhatsApp messages is converted into word embeddings using techniques such as Word2Vec, GloVe, or FastText, which map words into high-dimensional vector spaces while preserving semantic

Algorithm 1
Step 1. Input WhatsApp message containing emojis.
Step 2. Preprocess the message (tokenize and normalize emojis) and Feature Extraction using CNN
Step 3. Feed the pre-processed sequence to the trained BiLSTM model.
Step 4. Predict the emotional category of each emoji.
Step 5. Map the predicted emotions to corresponding text descriptions.
Step 6. Generate final output with emojis translated to text-based emotions.

relationships. Meanwhile, emojis—being non-textual elements—are encoded separately using one-hot encoding or emoji embeddings, where each emoji is assigned a distinct vector representing its emotional meaning. Once both text and emoji features are extracted, they are merged into a single feature matrix, allowing the model to analyse messages holistically.

Once the convolutional operation is performed, the output is a set of feature maps that highlight the presence and intensity of these detected patterns across the text. To further refine the representation, pooling layers such as max pooling are applied. Pooling reduces the dimensionality of the feature maps by retaining only the most significant values within each region. This step not only simplifies the data but also helps in preserving the most relevant features, reducing noise and redundancy.

Finally, we collect the obtained feature maps from all the filters in order to form a compact and high-dimensional representation of the input text. This concatenated feature vector encodes local contextual patterns and sentiment-rich information that is crucial to the later BiLSTM stage. With the CNN module in place, the system efficiently processes the text, emphasizing important emotional and contextual signals while conserving computational resources.

C.Emoji-Based Text Mining

This spectacular stage is the conversion of emoji's in WhatsApp messages into discernible text based on the emotional context, employing the Bidirectional Long Short-Term Memory (BiLSTM) algorithm in text mining. Emojis are extremely important in digital communication, as they set out an emotional context that integrates and often serves a simultaneous function-supplement of a textual expression. However, spreadsheet nature is the thing that encumbers sentiment analysis and natural language understanding. The suggested method uses the capability of BiLSTM to attain contextual dependencies from both past and future states within sequential data. WhatsApp message data is pre-processed using tokenization, emoji normalization, and stop word removal. Custom emoji embedding's merge with word to train the BiLSTM model, allowing emoji classification into predefined emotion categories such as happiness, sadness, anger, and surprise. The system generates text representations aligned with the emotional meaning of emoji's, improving sentiment analysis and user interaction applications. Finally, the extracted features are flattened and are passed through a fully connected network dense layer, where a softmax activation function classifies the message. The model assigns correspondingly positive, negative, or neutral sentiments to the text after classification.

Once the sentiment has been determined, the next stage is to map the detected sentiment onto appropriate emojis that visually depict the emotional context. For example, a positive sentiment can use emojis like 😀 (smile) or 😍 (heart eyes) to express excitement and happiness, negative sentiment would correspondingly use emojis like 😔 (disappointed face) or 😟 (angry face) for sadness and anger. While in neutral sentiments, an emoji such as 😐 (neutral face) can depict indifference or lack of strong emotion.

This can use preset conversions, whereby each sentiment category has a manual assignment to a set of emoji's, or learned conversions, where a model learns to pair specific emotional signals in the text to emoji's through training data. The goal is to enhance communication by visually conveying the sentiment behind the text, making interactions more expressive and engaging.

Table 3 illustrates how emojis are mapped to their corresponding emotional categories and how they are represented in text. Emotion Mapping are classified into three categories:

- **Positive Emojis** like 😀, 😁, ♡ are mapped to emotions of happiness, joy, or love.
- **Negative Emojis** like 😢, 😠, 😔 are mapped to emotions of sadness, anger, or disappointment.
- **Neutral Emojis** like 😐, 😑, 😶 represent emotions of indifference or neutrality.

After classifying the sentiment and mapping it to the corresponding emojis, the final step is to merge these emojis back into the original text to enrich the communication. Emojis are used to visually reinforce the emotional tone of the message, providing a clearer emotional context and enhancing user engagement. For example, in the sentence "The product is great 😍, but the delivery was late 😔," the words "great" and "late" are paired with the emojis 😍 (representing happiness or excitement) and 😔 (representing sadness or disappointment), respectively. This combination makes the message more dynamic, allowing the reader to instantly grasp the sentiment behind each part of the sentence. The use of emojis can help convey emotions more effectively than words alone, making the output feel more personalized and relatable. By integrating emojis with the text, the output not only becomes visually appealing but also

Table 3 Emoji-Mapped Text Representation

EMOJI	EMOTION CATEGORY	MAPPED TEXT REPRESENTATION	SAMPLE EMOTION EXAMPLE	EXAMPLE MESSAGE
😄	Positive	Happy	Feeling joyful or happy	"I'm feeling so great today! 😄 "
😂	Positive	Joy	Feeling excited or cheerful	"That was so funny! 😂 "
❤️	Positive	Love	Expressing affection or love	"I love this place! ❤️ "
😊	Positive	Content	Feeling content or satisfied	"Everything is going well 😊 "
😢	Negative	Sad	Feeling sorrow or sadness	"I lost my keys 😢 "
😡	Negative	Angry	Feeling angry or upset	"I'm really angry about this! 😡 "
😞	Negative	Disappointed	Feeling let down or disappointed	"That didn't go as expected 😞 "
😔	Negative	Upset	Feeling upset or sorrowful	"I'm feeling down today 😔 "
😐	Neutral	Neutral	Feeling neither happy nor sad	"Just another normal day 😐 "
😑	Neutral	Indifferent	Feeling indifferent or emotionless	"I don't have much to say 😑 "
😶	Neutral	Speechless	Feeling unable to express oneself	"I'm just lost for words 😶 "
😌	Neutral	Relaxed	Feeling calm or at ease	"Just enjoying the moment 😌 "

communicates the emotional nuances of the message in a more direct way.

These mappings can be useful in applications such as sentiment analysis, chatbot emotion understanding, social media analysis, and enhancing communication systems with a deeper understanding of emoji-based expressions

IV. RESULT AND DISCUSSIONS

The proposed model for Emoji to Text Conversion is evaluated by various performance metrics in terms of the effectiveness of the model concerning accuracy, precision, and efficiency. Here are several performance metrics that might apply:

A. *Accuracy*

The percentage of correct emoji predictions relative to the total number of predictions. It helps to determine how well the model is performing overall in mapping text to the correct emojis.

$$Accuracy = \frac{Number\ of\ Correct\ Prediction}{Total\ Number\ of\ Prediction} \times 100$$

B. *Precision*

The proportion of positive predictions (correct emoji classifications) out of all predictions made as positive. It's useful when the dataset has imbalanced classes, as it gives insight into how many of the predicted emojis are relevant. It Measures the accuracy of the emoji mapping when the model predicts a specific emoji for a given sentiment.

$$Precision = \frac{TP}{TP + FP}$$

C. *Recall*

The proportion of actual positive instances (correct emoji) that are correctly predicted by the model. Recall is important when it's critical not to miss any potential emoji classifications. It evaluates how well the model captures all the relevant emoji mappings.

$$Recall = \frac{TP}{TP + FN}$$

D. *F1 Score*

The harmonic mean of precision and recall, offering a balance between the two. It's especially useful when dealing with imbalanced classes. It Provides a single metric that balances precision and recall, indicating how well the model handles both false positives and false negatives.

$$F1\ Score = 2 \times \frac{Precision\ \times Recall}{Precision\ + Recall}$$

The comparison of the proposed method to the existing methods such as CNN, LSTM is given in Table 1.

Table 1 Comparison table for the Proposed Method

Model	Accuracy	Precision	Recall	F1-Score
CNN	0.78	0.81	0.75	0.78
LSTM	0.82	0.84	0.80	0.81
Proposed Method	0.85	0.88	0.84	0.86

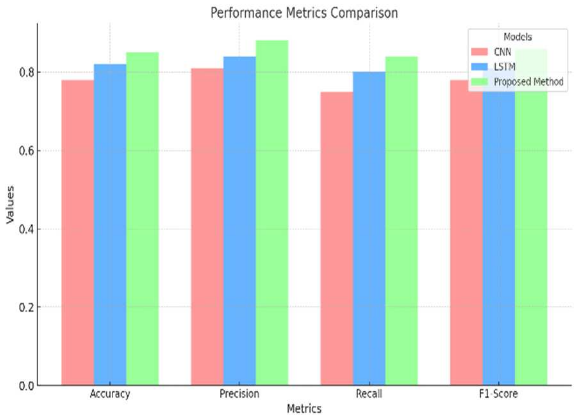

Fig 2 Comparison chart for Proposed Method

The Proposed Method outperforms both CNN and LSTM in all four metrics Accuracy, Precision, Recall, and F1-Score. Its superior performance can be attributed to its more advanced architecture or optimization techniques, making it better at both identifying and capturing relevant emojis.

V. Conclusion And Future Work

The proposed Text-Emoji Model has effectively enhances text mining for WhatsApp messages by integrating both textual and emoji-based sentiment analysis. By leveraging CNN for spatial feature extraction and BiLSTM for contextual learning, the model captures both local and sequential dependencies, leading to improved sentiment classification accuracy. The incorporation of word embeddings, emoji embeddings, and sentiment scores ensures a more comprehensive understanding of user emotions in digital communication.

Experimental results demonstrate that the proposed model outperforms traditional machine learning approaches in terms of accuracy, precision, recall, and F1-score, making it a robust solution for social media sentiment analysis, AI-driven chatbots, and customer feedback monitoring. The deployment of this model can significantly enhance human-computer interaction by enabling more emotionally intelligent and context-aware text processing systems. Future improvements may include attention mechanisms, transformer-based models (like BERT), and multilingual support to further enhance the model's performance and adaptability in real-world applications.

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